

Do Gaussian Splats Keep the Nuclei?

An Object-Level Audit of Gaussian Volume Compression for Fluorescence Microscopy

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Abstract—Gaussian splats are being adopted to store and stream large 3D microscopy and medical volumes, with budgets chosen and quality reported by PSNR. We ask whether, at a given number of bytes on disk, standard analysis still finds the nuclei. On four densely annotated frames from two *C. elegans* embryos (1,095 labelled nuclei), we compare a Gaussian-splat compressor (Luxar) with JPEG2000 and JPEG-XL at identical stored bytes, scoring a blob detector, a classical 3D watershed and a learned segmenter (Cellpose) against human labels, with paired confidence intervals and decision rules fixed in advance. At every size from $\approx 28\text{--}30\times$ to $530\times$, a correctly configured JPEG2000 has higher PSNR and, for both segmenters, keeps at least as many nuclei as the splats—significantly more in 23 of 30 (watershed) and 19 of 30 (Cellpose) size-matched pairs. Only the blob detector, which the splats’ smooth blobs suit, finds more nuclei in some splat reconstructions (by 3–4 points at $46\text{--}227\times$ on one embryo, and on an early frame whose nuclei are larger). Below ≈ 2 splats per nucleus the splats lose nuclei abruptly, while their PSNR stays within 19.1–22.6 dB. Gaussian splats are not a better storage format for this analysis than a standard wavelet codec; object-level, size-matched evaluation against a verified codec baseline should accompany such claims.

I. Introduction

Light-sheet and confocal time-lapse microscopy produce terabytes per experiment, and medical archives face the same pressure. Representing a volume as a set of anisotropic 3D Gaussians [1] is attractive for such data: it is compact, continuous and renders directly on the GPU. Recent systems fit Gaussian splats to microscopy volumes at scale [2], to slice-based microscopy and ultrasound [3], to medical volumes for compact storage [4], [5], and to tomographic reconstruction [6], [7]. Quality is measured by PSNR or SSIM throughout, and budgets are chosen by PSNR: Luxar selects its splat count by self-supervised held-out PSNR [8] and describes its fits as intended for visualisation rather than quantitative analysis [2].

What a microscopist needs from a compressed volume is that the analysis still works. Whether a nucleus remains detectable hinges on a small, often faint peak. Radiology learned this for lossy codecs, where acceptability is judged by whether lesions remain detectable, not by PSNR [9]. Gaussian volume compression has not been held to that standard, nor compared with standard codecs at identical stored bytes.

Contributions. (i) A rate-detectability protocol: identical bytes on disk, dense human labels, three detector

families, paired bootstrap intervals over nuclei, and decision rules committed before each new measurement. (ii) At identical bytes, JPEG2000 keeps at least as many nuclei as Gaussian splats for both segmenters at every size we tested, and loses only marginally to them under a blob detector at moderate ratios; Luxar’s own budget is safe, but so is the codec at the same bytes. (iii) Splat reconstructions cost a learned segmenter recall that blob detection does not reveal (a classical watershed loses less), and lose nuclei abruptly below ≈ 2 splats per nucleus with no PSNR signal. (iv) A baseline pitfall: a common JPEG2000 binding silently encodes a 3D array along the wrong axes, which had reversed our own high-ratio conclusion.

II. Related Work

Gaussian mixtures have long modelled nuclei for segmentation and lineage tracking [10]; our question concerns Gaussians as a storage format. Gaussian splatting [1] has been adapted to CT [7], [6], to slice-to-volume MRI with closed-form point-spread functions [11], and to volume compression [2], [3], [4], [5], each evaluated by image-fidelity metrics. JPEG2000 [12] and JPEG-XL [13] are the natural baselines at matched size. We use Cell Tracking Challenge labels [14], [15] and three detectors: a scale-normalised Laplacian of Gaussian [16], a distance-transform watershed, and Cellpose [17].

III. Methods

Data. Fluo-N3DH-CE [15]: *C. elegans* embryos imaged at $1.0 \times 0.09 \times 0.09 \mu\text{m}$ (8-bit, ≈ 12 MB per frame), with every nucleus labelled. We use embryo 1 (sequence 01) frames t150 and t194 and embryo 2 (sequence 02) frames t150 and t180, with 192 and 362 and 190 and 351 labelled nuclei; embryo 1 t100 (93 nuclei) enters only the budget analysis. Nuclei measure $3.6 \mu\text{m}$ (median over the challenge’s segmentation masks); all spatial parameters are fixed multiples of this diameter D .

Compressors. Luxar [2] with default settings, 1000 iterations and anisotropic voxel size, over seed budgets K from 10^3 to 6.4×10^4 (embryo 2: 3.2×10^4); stored size is its `.gsplats.zarr` on disk. For embryo 1 we also run Luxar’s own calibration—blind-spot held-out PSNR [8]—to obtain its recommended K^* . Luxar renders are mapped

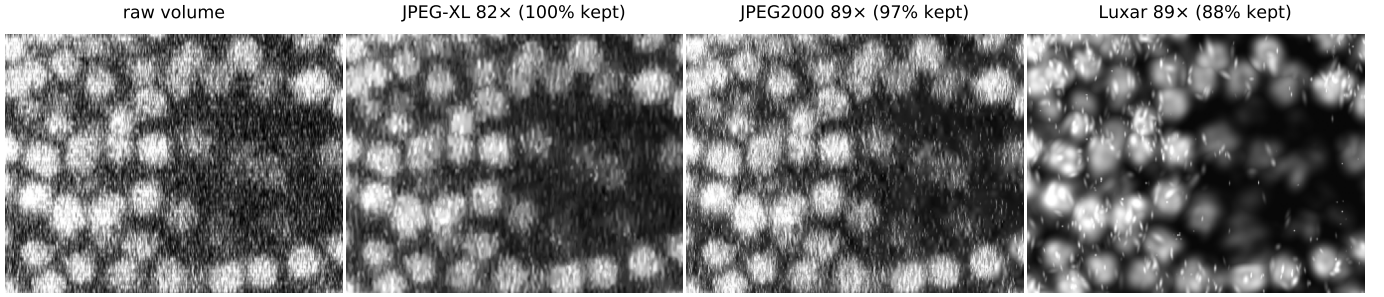


Fig. 1. The same region of embryo 1, t194 (3-slice maximum projection) from the raw volume, JPEG-XL, and JPEG2000 and Luxar at identical bytes; titles give whole-frame nuclei kept by Cellpose. Both codecs preserve the textured appearance of nuclei; Luxar renders smooth blobs on a black background with thin streaks from elongated splats.

to raw intensities by a least-squares gain and offset (two numbers, not counted in its size, which slightly favours it); its size does include the zarr metadata. JPEG2000 (OpenJPEG) codes each xy slice with a 2D wavelet, all slices as components of one codestream, and is bisected to exactly the byte size of every Luxar fit and to 50–800 KiB; JPEG-XL codes each slice and is bisected to 50–800 KiB.

A baseline pitfall. Passing a (z, y, x) array to a common JPEG2000 binding encodes a z - y image with one component per x column, so the wavelet never runs along x . That configuration, which we used at first, costs ≈ 10 dB at $220\times$ and made splats appear far better than JPEG2000 beyond $200\times$. We now check the codestream header (width, height, components) in a regression test.

Detectors and scoring. (a) LoG: local maxima in a physically isotropic window, ranked by scale-normalised LoG response ($\sigma \in \{0.2, 0.27, 0.33, 0.4\}D$); the top N are kept, N = number of labelled nuclei. (b) Watershed: background-subtracted, Otsu-thresholded foreground split by a distance-transform watershed in physical units; parameters fixed on raw volumes only. (c) Cellpose 3 nuclei, 2D per slice stitched across z . On the raw volumes they find 94%, 88% and 72% of the labelled nuclei. Detections match labels one-to-one within $0.6D$ (Table II varies it). Nuclei kept = nuclei found after compression / found in the raw volume by the same detector. Size-matched differences carry 95% bootstrap intervals over labelled nuclei, resampled jointly for the raw volume and both reconstructions; they are per pair, not corrected for multiple comparisons, so we report counts of pairs rather than single significant ones.

Decision rules. Before each new measurement we committed written pass/fail rules [TODO: repository link]; claims that failed are reported in their narrowed form.

IV. Results

Codecs. JPEG-XL is nearly lossless for detection up to its maximum distance (82–118 $\times$; worst case 92%). JPEG2000 continues far beyond: at 321–530 $\times$ it still keeps 100–102% (LoG), 97–100% (watershed) and 84–101% (Cellpose) of nuclei (Fig. 2).

TABLE I

Luxar minus JPEG2000 at identical bytes: nuclei kept (percentage points, mean over frames; superscript: fewer frames) and PSNR (dB). Bold: every pair's 95% interval lies on the same side of zero. Match radius $0.6D$.

Ratio	Splats/nuc.	Δ PSNR	LoG	Watershed	Cellpose
$\approx 30\times$	115–209	−6.7	+0.8 ²	−3.5 ²	−3.1 ²
$\approx 50\times$	57–115	−4.1	+1.4	−3.0	−10.5
$\approx 90\times$	30–56	−2.7	+1.0	−3.5	−7.5
$\approx 140\times$	16–30	−1.9	−0.2	−5.0	−11.8
$\approx 230\times$	5–13	−1.3	+0.3	−7.3	−10.1
$\approx 350\times$	2–6	−1.2	−6.3	−11.3	−12.1
$\approx 480\times$	1–2	−1.5	−24.6	−23.5	−25.2

Splats vs. JPEG2000 at identical bytes. Table I pairs every Luxar fit with the JPEG2000 encoding of the same size. For the segmenters, JPEG2000 keeps significantly more nuclei in 23 of 30 pairs (watershed) and 19 of 30 (Cellpose), and never fewer. Under the blob detector, Luxar keeps 3–4 points more at 46–227 $\times$ on E2 t150, E2 t180 (its smooth blobs act as a denoiser for a blob filter; kept exceeds 1), but beyond 300 $\times$ it keeps 66–101% against 100–102% for JPEG2000. On E1 t100, whose early-stage nuclei are larger than our fixed D , the blob detector's advantage on splats grows to 8–11 points at 159–537 $\times$, while neither segmenter favours Luxar in any pair there. JPEG2000 also has the higher PSNR in 30 of 30 size-matched pairs.

Luxar at its own budget. Luxar's calibration recommends $K^* = 64,000$ on both embryo-1 frames—the top of our grid, with held-out PSNR still rising. There it keeps 101% (LoG), 96–97% (watershed) and 97–98% (Cellpose) of nuclei, at 28–30 $\times$; JPEG2000 at the same bytes keeps 99–102%, at the cost of seconds rather than 17–46 min of fitting per volume on an A100 (12–27 min on an H100).

The budget cliff. Splats lose nuclei abruptly once they become scarce (Fig. 3): on all four frames Luxar drops below 90% (LoG) only at ≤ 2.3 splats per labelled nucleus, reaching 66–67% at one per nucleus, while its PSNR moves only within 19.1–22.6 dB. Survival—the fraction of the raw volume's 100 strongest LoG detections still found, which needs no labels—tracks nuclei kept across Luxar

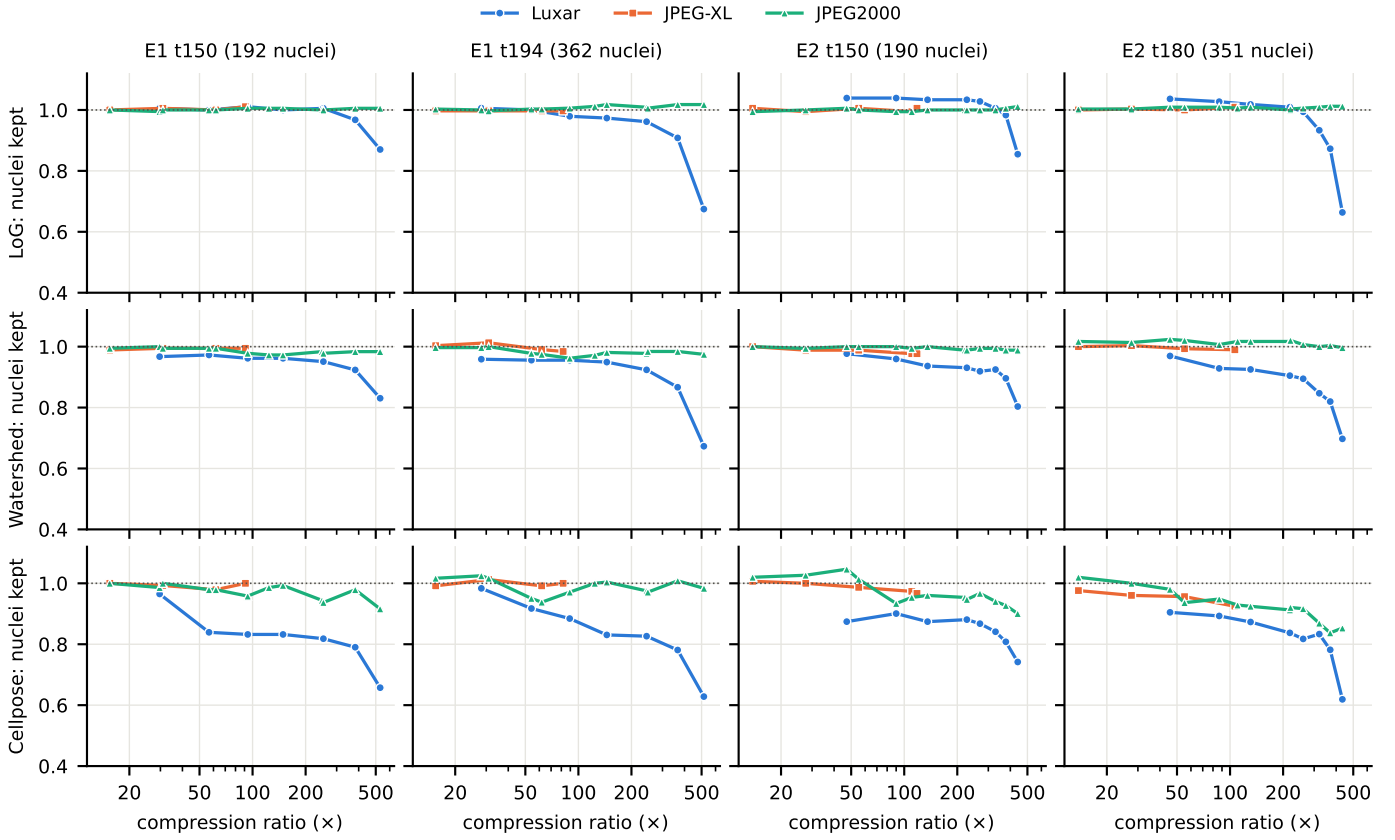


Fig. 2. Nuclei kept (relative to the raw volume, same detector) against compression ratio, for the LoG blob detector, the 3D watershed and Cellpose, on four frames from two embryos (E1, E2). JPEG2000 is encoded at each Luxar fit’s exact byte size and at 50–800 KiB.

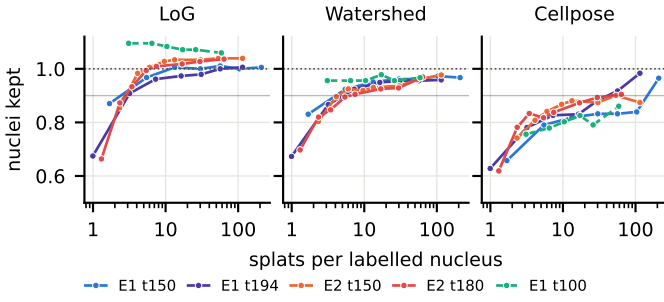


Fig. 3. The budget cliff: nuclei kept against splats per labelled nucleus for every Luxar fit, per detector (dashed: E1 t100).

fits (Spearman $\rho = 0.68$, $n = 30$) and can flag the cliff.

Detector dependence. At $45\times$ to $94\times$ the blob detector keeps 98–104% of nuclei in Luxar reconstructions, while Cellpose keeps only 83–92%: 6–15 points below JPEG-XL at the same ratios on all 4 frames (95% intervals exclude zero on 4 of 4). The classical watershed sits in between, at 93–98% and 2–4 points below JPEG-XL—consistently (intervals above zero on 4 of 4) but under the 5-point threshold we fixed in advance on three of four frames. The large penalty is thus specific to the learned segmenter: smooth blobs on a black background (Fig. 1) lie outside the

TABLE II
Match-radius sensitivity: nuclei found in the raw volumes (sum over four frames) and nuclei kept near $90\times$ (mean).

Radius	LoG			Cellpose		
	raw	JXL	Luxar	raw	JXL	Luxar
$0.4D$	1009	1.00	1.02	658	1.01	0.84
$0.6D$	1032	1.00	1.01	788	0.97	0.88
$0.8D$	1034	1.00	1.01	796	0.99	0.90

images it was trained on, whereas a blob filter is indifferent to them. Both segmenters nonetheless favour JPEG2000 at identical bytes, and the watershed’s deficit grows with compression (Table I). The blob detector recovers every nucleus once there are ≈ 3 –5 splats per nucleus; Cellpose needs tens to over a hundred, and on one frame never reaches 90% (Fig. 3). Cellpose’s scores move by up to 8 points between neighbouring sizes of the same codec, so single comparisons are noisy, but the sign is consistent across frames and embryos. Luxar itself is reproducible: three independent fits of E1 t194 at each of three budgets differ by at most 2.1 (LoG), 3.5 (watershed) and 3.7 (Cellpose) points of nuclei kept.

V. Discussion and Limitations

On this data, at identical stored bytes, Gaussian splats are not a better format for nuclear analysis than a standard wavelet codec: JPEG-XL is nearly lossless to $\approx 100\times$ and JPEG2000 keeps at least as many nuclei, for both segmenters, at every larger ratio we tested. Splats remain attractive for what Luxar designs them for—continuous, GPU-native rendering—and a blob detector tolerates them well. We recommend (i) comparing Gaussian volume compression with a verified codec baseline at identical bytes, (ii) reporting object-level recall with the detector that will be used downstream, and (iii) checking a splat budget at the object level before trusting it (survival is a label-free start).

Limitations: one organism (two embryos, five frames); Cellpose in its 2D-stitched mode; one Gaussian compressor at default settings (tuned or entropy-coded splats may fare better); a 2D-per-slice JPEG2000 (a 3D wavelet would likely be stronger still); Luxar’s recommended budget scored on embryo 1 only (its calibration on embryo 2 also recommends $K^* = 64,000$); and larger early-embryo nuclei (t100) that fit our fixed D less well. The same audit for small low-contrast lesions in CT is future work.

VI. Conclusion

Measured at identical bytes against human labels, Gaussian-splat compression keeps nuclei well at its own budget but no better than JPEG2000 at any budget for segmentation, loses them abruptly below a budget PSNR cannot identify, and costs a learned segmenter recall that a blob detector does not show. Object-level, size-matched evaluation against verified codec baselines should accompany Gaussian volume compression.

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